

Kunal Maheshwari

Haldwani, Uttarakhand, India | 9258446350 | work.kunal.maheshwari@gmail.com

LinkedIn: linkedin.com/in/kunalmaheshwari26 | GitHub: github.com/preferablehuman | Portfolio: preferablehuman.github.io/portfolio/

Summary

Python and AI Backend Developer with 4.6+ years of enterprise backend experience and graduate AI/cloud projects across FastAPI, PyTorch, Hugging Face Transformers, T5, PEFT/LoRA, ROUGE evaluation, Docker, MinIO, Prometheus, Java/Spring, REST APIs, SQL, and AWS/Azure integrations. Built privacy-oriented model-serving workflows, production healthcare APIs processing 50K+ daily requests, and data pipelines using Spark, Terraform, and cloud services.

AI and Backend Projects

Privacy-Preserving AI Email Client

FastAPI, T5, LoRA, PyTorch, Hugging Face

- Built project-based NLP model-service workflows for email summarization, categorization, and reply suggestions using Python, FastAPI, PyTorch, Hugging Face Transformers, T5, PEFT/LoRA, and structured output parsing.
- Implemented training/inference patterns with checkpoint recovery, ROUGE evaluation, Docker/MinIO artifacts, Prometheus metrics, CPU/GPU diagnostics, and local/privacy-oriented model lifecycle boundaries.
- Evaluated summarization outputs with ROUGE-1 F 53.04, ROUGE-2 F 44.90, and ROUGE-L F 45.90 while avoiding unsupported production-user or formal-compliance claims.

NYU Taxi Data Analysis Platform

Python, PySpark, AWS, Terraform

- Built AWS event-driven data platform using S3, Lambda, EFS, ECS Spark Structured Streaming, RDS PostgreSQL, Terraform, Docker, and CloudWatch, processing 8GB+ data with 40% runtime improvement.

Work Experience

NTT Data, Bengaluru, India

Software Developer, Aug 2022 - Aug 2024

- Developed Java 8/11, Spring Boot/Spring Framework, REST/SOAP, SOA, and Azure-integrated healthcare APIs, sustaining 50K+ daily requests and 99.99% SLA availability.
- Designed Azure Relay, Azure Service Bus, Azure Functions, JMS, API Gateway authentication, and database-propagation workflows; improved traceability with Log4j 2.x and reduced issue-resolution time 20%.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Built automotive Java/Spring backend services and integrated a Python decision-tree REST service for vehicle option availability, pricing, and business-rule validation.
- Improved production performance approximately 90% through Oracle SQL tuning, query refactoring, and legacy Java modernization while supporting Jenkins, GitLab CI/CD, Maven, and JUnit4 quality gates.

Technical Skills

AI/ML: PyTorch, Hugging Face Transformers, T5, PEFT, LoRA, ROUGE, Sequence-to-Sequence Models, Model Evaluation, Local LLMs, Model Lifecycle.

Python Backend: Python, FastAPI, Flask, Pandas, NumPy, SQLAlchemy, REST APIs, Structured Outputs, Docker, MinIO, Prometheus, OpenCV.

Cloud/Data: AWS S3, Lambda, EFS, ECS, RDS, CloudWatch, Terraform, PySpark, Spark Structured Streaming, PostgreSQL, Google Cloud Storage, Cloud SQL.

Enterprise Backend: Java 8/11, Spring Boot, Spring Framework, Spring MVC, REST/SOAP APIs, JMS, Azure Service Bus, Azure Relay, Jenkins, GitLab CI/CD, Maven, JUnit, Log4j 2.x.

Education and Certifications

Central Michigan University, MS Computer Science, Aug 2024 - Dec 2025, GPA: 3.87/4.0 | **Lovely Professional University**, B.Tech CSE, 2016 - 2020, CGPA: 8.02/10

Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate